
RE: Iowa DOT Streetlight Volumes and RAMS AADT for STATCOM

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Cc statcom@iastate.edu <statcom@iastate.edu>; Pedersen, Garrett <Garrett.Pedersen@iowadot.us>; Zamora, Jorge <Jorge.Zamora@iowadot.us>

 5 attachments (8 MB)

Revised_Guidelines_for_AADT_Estimates_Final_07082025_508 complete.pdf; 2024_streetlight_aadt_analysis.pdf; Texas_A&M_study_for_MNDOT.pdf; AADT 2024 Initial Release Methodology and Validation White Paper.pdf; StreetLight Data_Methodology and Data Sources.pdf;

Good Afternoon,

The following is the additional info that we discussed during our meeting, as well as some detail about the data that Justin shared with you this week. One item of note is that you'll see different volume ranges that data has been put into and reported on. This is the DOT count, and the FHWA source examples or Minnesota DOT real-world application would be good examples for you to use, but if you find that they could be improved on, we are open to suggestions.

a. Guidelines for Obtaining AADT Estimates from Non-Traditional Sources.

- a. Abstract – “This document provides decision making guidance that State DOTs and local transportation agencies can use to plan and execute purchases of traffic volume estimates, such as annual average daily traffic (AADT), from the private sector when those estimates are based on non-traditional data sources, such as vehicle probe or smartphone data. It also provides guidance for validating the quality of such estimates.”
- b. This is the update to the original study report.
- c. The table of contents items of likely significance
 - i. Introduction, page 1
 - ii. What is AADT and how it is measured, page 1
 - iii. Advantages and Limitations of AADT Estimates from Non-Traditional Sources, page 3
 - iv. A.3) What data quality requirements must the data meet and how will these be determined?, page 6
 - v. C.7) Follow accuracy testing procedures that are shown in Appendix B or audit the vendor's cross-validation analyses used for model development and testing for your state, page 33
 - vi. Appendix B: How to determine AADT accuracy and precision
 - vii. I believe the “Validation in cases with only portable count data” starting on page 61 is relevant (the prior appendix content is focused on the permanent locations, which we are not looking to replace at this time)
 - viii. Appendix D: Literature review on AADT estimation from non-traditional data (Big Data)

b. Evaluation of StreetLight Data's Traffic Count Estimates from Mobile Device Data.

- a. This is a TTI analysis of the Minnesota DOT effort of using StreetLight data from November 2020
- b. This is written with the first draft of the FHWA guidance, rather than the above second version that has been described

- c. The report covers permanent and short-duration count results
- d. The document includes a recommended review process
- e. Items of likely significance
- f. Page 9 of 24, shows the recommendations
- g. Pages 2-4 of 24, shows the evaluation methods
- c. **2024 MnDOT Mobile Source Data for AADT Implementation Analysis**
 - a. This is a review of the Minnesota use of StreetLight, building upon the 2020 work listed above of item “b”
 - b. Documents the results of expanded into a new, lower volume group.
- d. **StreetLight Data_Methodology and Data Sources**
 - a. This reviews the overall process that the company uses to product traffic estimates
 - b. Page 4 of 12, Figure 1 shows how smart phone application (yellow) has decreased as a source, and in-car (blue) and contextual data (gray) has increased to inform how data can be translated into traffic estimates.
 - c. Page 6 of 12, shows an image and text on the locking of data onto to roadways, that can then be used to model/report trips, patterns, etc.
 - d. Onwards from there is just good info on expansion, validation
- e. **AADT 2024 Initial Release Methodology and Validation White Paper**
 - a. This talks about the practice in a more general manner.
 - b. Page 4 of 18, shows a map of what permanent counters are used in validation. In the past we have seen that they at least tested against Iowa data. Perhaps Iowa data is used in the past or in 2025, but that is not known for sure. It is possible that they use the short-term count data instead, but that is not known for sure either.
 - c. Page 6 of 18, shows short-term count error targets, we have not looked to see how this compares to the FHWA guidance or MN example.
 - d. Page 9 of 18, shows their estimation of predictive power of their data. We did not see results this good from the Battelle review, but the data that Justin supplied showed a very strong r-square.

Per the data that was provided to you through Dr. Nettleton, here is a header explanation for the different tabs, we have or will be supplying a new data set of the interstate only links, so this is only explanatory for how Battelle analyzed our data

- Tab: “Data” – the conflated dataset supplied to
 - station_id – the DOT unique identifier
 - locality – the determination of land use as a variable, rural and urban
 - aadt_jack – this is the DOT processed count (our current practice output)
 - aadt_sld – this is the StreetLight estimate
 - tce – traffic count error, which is the difference between the Streetlight and DOT figure
 - atce – the absolute calculation of the traffic count error
 - vol_range – FHWA established ranges for the binning of data
 - tce_range – I assume that is part of the statistical test shown here
 - within_range – an indicator whether a streetlight count
- Tab: “Comb Results Table” – this is a summary from Battelle, the color fill is used to show when a test is passed (green) or failed (red)
- Tab: “Summary” - is a binned results table
- Tab: “Summary for analysis” – similar to the above with some lines added
- Tab: the rest are specific graphs for results by combinations of year, the test, and land use.

Per the dataset that was sent by Justin this week, here are some additional notes for consideration. Of course if you have questions, please let us know.

- General notes
 - We know that StreetLight takes some of our data, that is publicly available, and uses it to train their model in the estimation of AADT. The danger of moving to using their data, is that then there would no longer be the DOT data to use for training, and accuracy could suffer. We’d really like to know if you can determine how independent their data is from ours, as at initial glance, the r-squared for all data is very close (~0.97)
 - We have included counts for the motorway and ramps
- Folders/Files
 - CONFLATED
 - This is a subset of the two datasets, were we determined the selected links were conflated if there is a 98% overlap between.

- We can decrease the % to get more links, but the dataset has several hundred ramp and motorway links. If you have a better method for conflation of more links, we'd be happy to see what you can get.
- Data
 - There are various unique identifiers, but the "ID" would be our suggestion
 - To differentiate between mainline roadway links and the ramps the field "Zone Name" has;
 - "Motorway_link" is used for ramps
 - The route name for mainline links is unique
 - AADT_IADOT, is the DOT count
 - VOLUME_STL, is the StreetLight estimation
- RAMS_EXPORT
 - This has similar but not all fields, the common field from this layer to the CONFLATED layer is "ID"
- STL_EXTRACT
 - In this folder, there is 2022 and 2024 data, and for both there are motorway and ramp files.
 - This has similar but not all fields, the common field from this layer to the CONFLATED layer is "segment_id"

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From: Meade, Justin <Justin.Meade@iowadot.us>
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Subject: Iowa DOT Streetlight Volumes and RAMS AADT for STATCOM

Sam, Liam, and STATCOM,
 I've exported data from RAMS and Streetlight Insight.
 The data exports are attached to this email.
 I've conflated the AADT from the RAMS network to the OSM segments provided by Streetlight.
 I only conflated segments when I could be very certain that the segments should be spatially related.

The FME workspace that managed the conflation and exported the RAMS data is included as well.

Please let me know if there is anything else that your team needs and apologies if I left anyone out!
 These were all the email accounts attached to the December 3rd meeting.
 Thank you!

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